

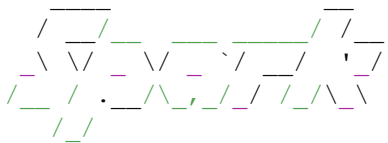
✓ PRACTICAL – SPARK (As per your PDF)

PART 1 – STARTING SPARK SHELL

```
spark-shell
```

Expected Output (similar to your PDF):

Welcome to

The Spark logo is a stylized representation of a star or a cluster of points, composed of several colored lines (green, yellow, red, blue) that form a geometric shape. It is positioned to the left of the text "version 1.6.0".

version 1.6.0

Using Scala version 2.10.5 ...

PART 2 – SIMPLE RDD FROM A SCALA LIST

STEP 1 – Create a sequence of strings

```
val data = Seq("good morning", "good morning everyone", "good morning all")
```

STEP 2 – Create an RDD from it

```
val myRdd = sc.parallelize(data)
```

STEP 3 – Display entire RDD

```
myRdd.collect.foreach(println)
```

Expected Output

```
good morning
good morning everyone
good morning all
```

STEP 4 – Display only first 2 elements

```
myRdd.take(2).foreach(println)
```

STEP 5 – Convert RDD to DataFrame

```
myRdd.toDF().show()
```

Expected Output

```
+-----+
| value          |
+-----+
| good morning   |
| good morning eve... |
| good morning all   |
+-----+
```

PART 3 – WORD COUNT USING RDD

STEP 1 – Split lines into words

```
val wordsRdd = myRdd.flatMap(line => line.split(" "))
```

STEP 2 – Display words

```
wordsRdd.collect().foreach(println)
```

Expected sample output:

```
good
morning
good
morning
everyone
```

```
good
morning
all
```

STEP 3 – Convert each word to (word,1)

```
val mapRdd = wordsRdd.map(word => (word, 1))
```

Check output:

```
mapRdd.collect.foreach(println)
```

Example:

```
(good,1)
(morning,1)
(everyone,1)
(all,1)
```

STEP 4 – Reduce to count all words

```
val reduceRdd = mapRdd.reduceByKey(_ + _)
```

Show results:

```
reduceRdd.collect.foreach(println)
```

Expected output

```
(good,3)
(morning,3)
(everyone,1)
(all,1)
```

PART 4 – WORD COUNT USING A TEXT FILE

STEP 1 – Load text file from HDFS

```
val txtrdd = sc.textFile("/user/cloudera/input.csv")
```

STEP 2 – View content

```
txttrdd.collect.foreach(println)
```

Example output from PDF:

```
car bear river  
river car river  
bear bear car  
car car
```

STEP 3 – Split into words

```
val txtwordsRdd = txttrdd.flatMap(line => line.split(" "))
```

Show words:

```
txtwordsRdd.collect.foreach(println)
```

STEP 4 – (word,1) mapping

```
val txtmapRdd = txtwordsRdd.map(word => (word,1))
```

Show:

```
txtmapRdd.collect.foreach(println)
```

STEP 5 – Reduce word counts

```
val txtreduceRdd = txtmapRdd.reduceByKey(_ + _)
```

Show final output:

```
txtreduceRdd.collect.foreach(println)
```

Expected Sample Output

```
(car,7)  
(bear,3)  
(river,4)
```